GRADUATION PROJECT / PORTFOLIO SUMMARY

Plant Design and Site Selection

Production of n-Butanol from Ethanol

I completed this project with Yousef Rakan Freih.
Chemical Engineering, King Saud University | Graduation Project 2 (CHE 496)

50,000	97%	2	8
tonnes/year design target	product purity target	industrial sites compared	site selection criteria

MY AIM

I wanted to assess whether producing n-butanol from ethanol could be feasible and whether Jubail or Yanbu would be the better site. I worked with my project partner on process design, cost estimates, financial calculations and a safety review.

WHAT I WORKED ON

- **Process design:** I helped prepare the process flowsheet, calculate material and energy balances, and size the reactor, heat exchanger, separator, distillation column and pumps.
- **Costs and profitability:** I worked on equipment and investment cost estimates, reviewed cash flows, and tested how +/-20% changes in four inputs affected the financial results.
- **Safety and the environment:** I helped review hazards using HAZOP and consider controls for feed handling, reaction, separation, storage and wastewater treatment.
- **Site selection:** I compared Jubail and Yanbu using eight factors, including raw material access, utilities, transport, land and industrial services.

WHAT I CONCLUDED

I supported Jubail as the preferred site in the study. I connected the process and equipment needs to costs, utilities and risks when comparing the sites. The results suggested that the project was worth further study, but I would need current prices and checked financial figures before recommending an investment.

Tools used in the project: POLYMATH for reactor calculations; CAPCOST and bare-module costing for equipment cost estimates.
I have included the detailed calculations, assumptions and references in the full project report.

Costs and Site Selection

My financial results and comparison of Jubail and Yanbu

FINANCIAL ESTIMATES FROM THE PROJECT

US$16.3m
estimated total investment

US$0.78m
estimated net present value

13.54%
estimated discounted cash-flow return

I used these estimates in the academic study. They are not actual plant returns. I would check the financial differences on page 3 before using them for a business decision. The amounts above are in US dollars.

WHICH INPUTS AFFECTED THE RESULTS MOST

I tested four inputs using +/-20% changes. I found that **butanol selling price** had the biggest effect, followed by **utility cost**, **raw material cost** and **total investment**. I would check these inputs first when updating the analysis.

HOW I COMPARED THE TWO SITES

Criterion	Jubail	Yanbu
Raw materials	10	8
Utilities	10	9
Transportation	10	9
Land and expansion	10	8
Safety and industrial services	9	9
Labor availability	9	9
Export logistics	10	9
Weather suitability	7	7
Total score	**75**	**68**



I have kept the project scores: eight criteria scored out of 10, giving a total of 80. I have not added any extra weights to the scores.

WHY I SUPPORTED JUBAIL

I supported Jubail because the study showed better access to raw materials and hydrogen, utilities, transport and space for expansion. It scored **75 out of 80**, compared with **68 for Yanbu**. I would confirm current prices and site services before using this result for an actual plant.

Risks and Next Steps

What I found and what I would check next

MAIN RISKS AND MY RECOMMENDATIONS

Prices and costs

I found that changes in selling price and input costs could change the financial result. I would check customer prices, supplier terms and utility charges before relying on the estimates.

Process and hydrogen safety

I helped review risks from high temperatures, high pressure and leaks. I would check backup sensors, pressure relief, emergency shutdowns and spacing between equipment.

Energy use and separation

I reviewed how distillation and heat transfer affect cost and product quality. I would check options to reuse heat and improve separation before estimating any savings.

FINANCIAL FIGURES I WOULD CHECK

I found different payback figures in the full report. Discounted payback is **3.2 years** in the profitability summary and about **5.3 years** in the cash-flow discussion. Payback without discounting is given as **2.5** and **4.5 years** in different sections. I would check the calculations, timing and cash flows to resolve these differences.

I also noticed that a utility-cost table is headed AED while other financial sections use dollars. I would confirm the currency and put the costs and cash flows on the same basis. I have not selected a payback figure or converted currencies in this summary.

MY NEXT STEP

I would keep **Jubail as the preferred site for further study**. Next, I would get updated supplier and market prices, confirm the production rate and construction timing, correct the currency differences, and recalculate the returns. I would then repeat the price and cost tests before recommending an investment.

WHAT I LEARNED

I learned how to connect process calculations with costs, compare two locations, and explain the risks behind a financial result. I also learned to check whether the numbers agree before making a recommendation.

MY SOURCE REPORT

I prepared this summary from Production of n-Butanol from Ethanol, which I co-authored with Yousef Rakan Freih at King Saud University for Graduation Project 2 (CHE 496), first semester 1447 H / 2025 G. The full report has 119 PDF pages.

Where to find the details (printed page numbers): problem statement p.2; design basis p.12; equipment and economics pp.85-97 (profitability summary p.92); site comparison pp.106-107; recommendations pp.113-114. Add one to each printed page number to find its PDF page.

Scope: I worked on an academic plant design and feasibility study. I would need operating data to measure any cost or environmental improvements.